Course	BSCS	Academic Year	2026-2027
Activity	Research Topic Discovery & Regrouping	Time	60-90 minutes
Class Organization	Individual to Interest Groups	Output	Group Research Concept
References	UN SDGs + DOST CALABARZON	Assessment	100 points

Activity Overview

This activity prepares students for research group formation. Students identify a research interest, connect it to a United Nations Sustainable Development Goal (SDG), explore a relevant DOST CALABARZON Hack4aProgress challenge track, and form a group around a common research problem.

This completed research concept focuses on the Web-Based Digital Parking Validation System for Laguna State Polytechnic University Santa Cruz Campus. It follows the features described in the project proposal.

Reference websites: UN Sustainable Development Goals - https://sdgs.un.org/goals
DOST CALABARZON Hack4aProgress - https://hack4aprogress.dostcalabarzon.ph/

Learning Objectives

• Identify a school or community problem that can be investigated.

• Relate the problem to an appropriate SDG.

• Examine the closest Hack4aProgress challenge track and state any limits in alignment.

• Form a research group based on common research interests.

• Develop a preliminary title, problem, research questions, and technology or data approach.

Part I - Explore the SDGs

Visit https://sdgs.un.org/goals and review the 17 Sustainable Development Goals. Choose ONE SDG that interests you most.

My Chosen SDG Number	9
SDG Title	Industry, Innovation and Infrastructure
Why did I choose this SDG?	I chose SDG 9 because the study applies digital innovation to improve an existing campus process. A web-based validation system can support more reliable parking registration, verification, and record management.
What problem related to this SDG interests me?	Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage.

Who is affected by this problem? (Check all that apply)

[X] Students [X] Teachers [ ] Farmers [ ] Fisherfolk [ ] Families
[ ] LGUs [ ] Businesses [X] Communities [ ] Environment
[X] Other: Campus staff, security personnel, and parking administrators

Part II - Explore Hack4aProgress

The challenge tracks below are reproduced from the reference activity.

Track 1	AI for Agriculture & Food Security
Track 2	AI for Energy Resilience
Track 3	AI for Disaster Resilience
Track 4	AI for Environmental Sustainability
My chosen track	Closest thematic track: Track 4 - AI for Environmental Sustainability. The proposal reduces paper use but does not include AI; direct alignment with an AI challenge is not established.

Why did you choose this track?

Replacing printed stickers may reduce paper use, giving the proposal a limited sustainability connection. However, its core purpose is parking validation and campus record management. No AI functionality is claimed or added to the proposed scope.

Part III - Connect the SDG + Local Problem + Technology

SDG	9 - Industry, Innovation and Infrastructure
Hack4aProgress Track	Closest thematic track: Track 4 - AI for Environmental Sustainability. The proposal reduces paper use but does not include AI; direct alignment with an AI challenge is not established.
Local/Community Problem	Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage. The study focuses on this risk at LSPU Santa Cruz Campus; its frequency must be established during requirements gathering.
Target Beneficiaries	Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.
Possible Technology/Approach	[ ] AI/ML [ ] Data Analytics [ ] GIS [ ] Mobile App [X] Web App [ ] IoT [ ] Computer Vision [X] Other: Secure online database and digital validation
Initial Research Idea	Develop a web-based system that allows vehicle owners to register their vehicles and receive digital parking passes, security personnel to verify records digitally, and administrators to monitor and manage parking registrations.

Part IV - The Study That Interests Me the Most

Transform the general interest into a researchable study.

Preliminary Research Title	Web-Based Digital Parking Validation System
Research Problem	Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage.
Research Question 1	What registration, verification, and record-management requirements do vehicle owners, security personnel, and parking administrators have?
Research Question 2	How can a web-based system support vehicle registration, digital parking pass generation, verification, and administrative monitoring?
Research Question 3	How does the developed system perform in terms of functional correctness, usability, verification time, and rejection of invalid or unauthorized records?

Part V - REGROUP YOURSELF

Find classmates who share a similar SDG, school problem, target beneficiary, technology, or research direction. Form a group based on research compatibility.

Suggested discussion questions

• What SDG did you choose? SDG 9 - Industry, Innovation and Infrastructure.

• What problem do you want to investigate? The risk of copied parking stickers and the difficulty of manually verifying and managing campus parking registrations.

• What Hack4aProgress track did you choose? Track 4 is the closest thematic connection through reduced paper use. The proposal has no AI component.

• Who are your target beneficiaries? Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.

• What technology could address the problem? Web application, secure online database, digital parking passes, digital verification, and an administrative dashboard.

Research Interest Card

Name	
SDG	SDG 9 - Industry, Innovation and Infrastructure
Problem	Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage.
Hack4aProgress Track	Closest thematic track: Track 4 - AI for Environmental Sustainability. The proposal reduces paper use but does not include AI; direct alignment with an AI challenge is not established.
Target Beneficiary	Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.
Technology	Web application, secure online database, digital parking passes, digital verification, and an administrative dashboard.
Possible Research Title	Web-Based Digital Parking Validation System

Part VI - Group Research Team

After regrouping, agree on ONE research direction.

Member	Research Interest	Possible Role
1.		
2.		
3.		

Suggested roles: Research Leader, Problem Analyst, Data Researcher, Technology Researcher, Documentation Lead.

Part VII - Group Research Concept

Group No.	
Proposed Research Title	Web-Based Digital Parking Validation System
SDG Alignment	SDG 9 - Industry, Innovation and Infrastructure. The study introduces a digital alternative to printed parking stickers and supports campus registration, verification, and record management. Reduced paper use is a secondary benefit.
Hack4aProgress Alignment	Closest thematic track: Track 4 - AI for Environmental Sustainability. The proposal reduces paper use but does not include AI; direct alignment with an AI challenge is not established.
Research Problem	Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage. The proposed study will determine campus requirements and evaluate a digital alternative.
Target Beneficiaries	Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.
Proposed Research Approach	[X] Survey [ ] Experimental [ ] Data Analysis [ ] AI/ML [X] System Development [ ] GIS [X] Evaluation Other: Requirements gathering and functional testing
Possible Data Needed / Sources	Vehicle registration details and validation status; existing parking registration procedures and records, subject to authorization; requirements from vehicle owners, guards, and administrators; test results; task completion times; and user evaluation responses.
Expected Output	A functional web-based parking validation system with vehicle registration, digital parking pass generation, a secure online database, digital verification for security personnel, and an administrative dashboard, together with system documentation and evaluation reports.

Part VIII - Research Title Formula

Use this guide:

Development / Assessment / Analysis / Prediction of + WHAT + FOR WHOM + WHERE + USING WHAT

Application to the proposed study: Development of a digital parking validation system for vehicle owners, security personnel, and parking administrators at LSPU Santa Cruz Campus using web technology and a secure online database.

Proposed Research Title

Web-Based Digital Parking Validation System

Study Site: Laguna State Polytechnic University Santa Cruz Campus

Part IX - 2-Minute Research Idea Pitch

1. What is the problem?

Printed parking stickers can be photocopied or reproduced, making it difficult for security personnel to distinguish valid registrations from fake stickers. Manual checking also makes parking records harder to manage.

2. Who is affected?

Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.

3. Why is the problem important?

Reliable verification helps security personnel distinguish valid registrations from invalid records. Digitizing the process can make registration more convenient, improve record management, and reduce reliance on printed stickers.

4. What SDG does the study address?

SDG 9 - Industry, Innovation and Infrastructure, through digital innovation in campus parking validation.

5. What Hack4aProgress track does it address?

Track 4 has the closest thematic link through reduced paper use. The current proposal is a web-based information system and does not directly address an AI challenge.

6. What do you propose to investigate or develop?

We propose to develop and evaluate a web-based digital parking validation system for LSPU Santa Cruz Campus. Vehicle owners will register online and receive digital parking passes. Security personnel will verify the corresponding validation records, while administrators will manage registrations through a dashboard. Evaluation will cover the required functions, usability, verification time, and handling of invalid records.

Final Group Output - Research Concept Canvas

Research Title	Web-Based Digital Parking Validation System
SDG	SDG 9 - Industry, Innovation and Infrastructure
Hack4aProgress Track	Closest thematic track: Track 4 - AI for Environmental Sustainability. The proposal reduces paper use but does not include AI; direct alignment with an AI challenge is not established.
Community Problem	Risk of copied parking stickers and inefficient manual verification and parking record management at LSPU Santa Cruz Campus.
Target Beneficiaries	Students and staff who drive to campus, security personnel, and the school administration or parking office at LSPU Santa Cruz Campus.
Research Questions	1. What registration, verification, and record-management requirements do vehicle owners, security personnel, and parking administrators have? 2. How can a web-based system support vehicle registration, digital parking pass generation, verification, and administrative monitoring? 3. How does the developed system perform in terms of functional correctness, usability, verification time, and rejection of invalid or unauthorized records?
Possible Data	Registration requirements, authorized parking records, stakeholder feedback, functional test results, verification times, and usability responses.
Proposed Technology	Web application, secure online database, digital parking passes, digital verification, and an administrative dashboard.
Expected Output	Functional web-based parking validation system, secure database, digital passes, verification interface, administrative dashboard, documentation, and evaluation reports.
Group Members	

Assessment Rubric

Criteria	Points	Score
SDG Alignment	15	
Problem Identification	20	
Relevance to Community	20	
Hack4aProgress Alignment	15	
Researchability / Feasibility	15	
Creativity / Innovation	10	
Group Participation	5	
TOTAL	100	